01 JONATHAN

Jonathan is a participant at Pegasus, an organization for adults with developmental disabilities. Recently, he has taken an interest in filming and operating the community's GoPro camera. However, this proves to be a challenge for the following reasons:

01 Jonathan struggles to **hold** the 200g camera, for a prolonged period of time

02 He places the camera on his lap, which makes it **difficult to see** the screen

03 Without comfortable visibility, Jonathan has **little control** in the camera **movement**.

Hold

See

Move

02 OPPORTUNITY

We, the design team, want to help Jonathan operate the camera as he envisions. This means he must be able to **see** and **move** the camera without **holding** it (constraint).

01 **Visibility**

02 **Mobility**

05 NEXT STEPS

For Jonathan's opportunity, we strongly recommend our wheelchair-mounted RC filming gimbal. The immediate next step is real-world implementation and feedback from Jonathan.

Areas of Improvement:

- Comparing latency of the monitor vs a tablet implementation
- Iterating on the force necessary for the push clamps
- Building and testing current prototypes

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WRC FILM

Wheelchair-Mounted RC Filming Gimbal

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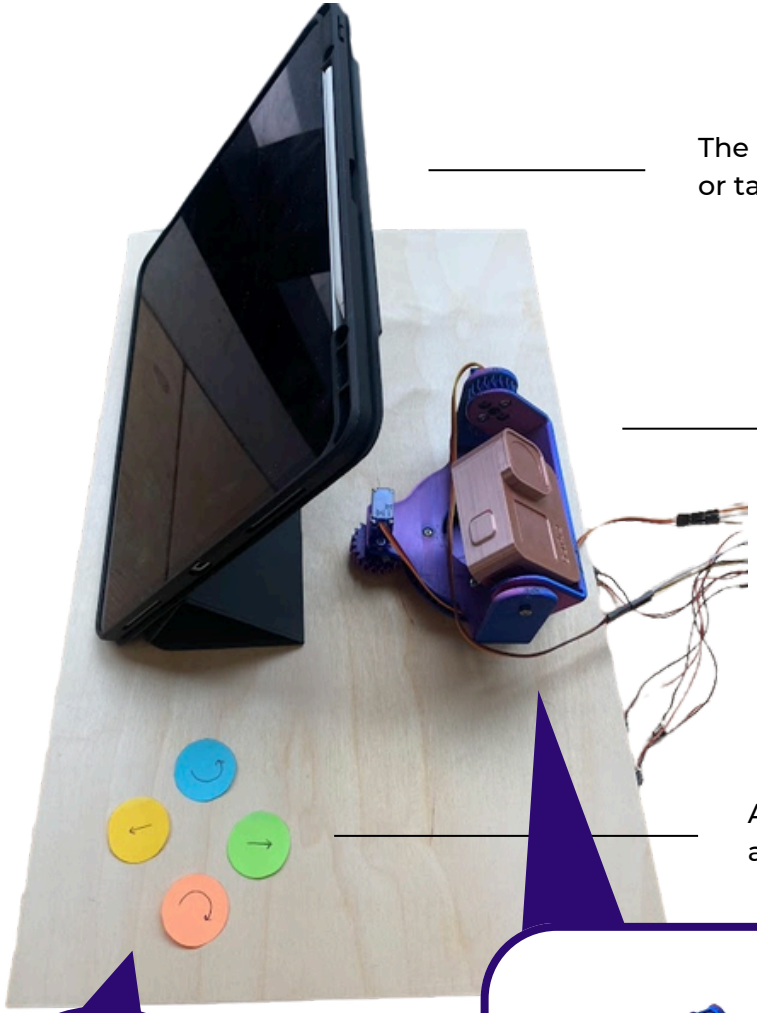
The remote-control gimbal allows for both rotation and tilt with minimal physical strain. This design will enable Jonathan, a wheelchair user with difficulty holding up a camera, to film independently as he envisions.

03 OBJECTIVES

- Remove the need for continuous holding
- Maintain a 35 cm viewing distance at eye-level
- Allow for smooth rotation and tilt

04 SOLUTION

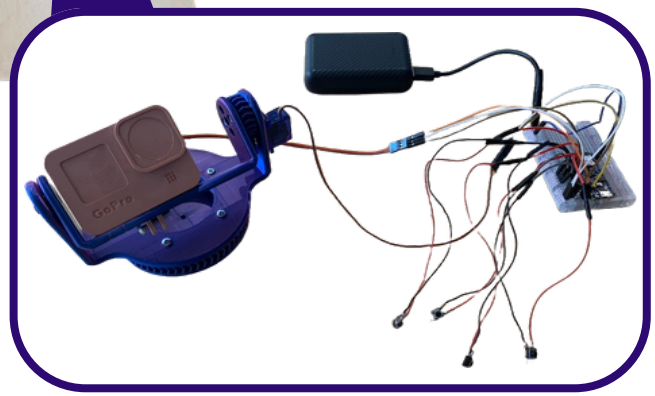
- The table-like surface removes the need for continuous holding
- The monitor/tablet feature allows for adjustable viewing distance
- Two servos allow for rotation and tilt
- An intuitive button design allows for easy control



The camera can be connected to a monitor or tablet, allowing for easy viewing

Two servos enable smooth 360° rotation & 108° tilt

A button pad designed for accessibility allows for easy control



Not Pictured:
Push clamps attached to the bottom of the table ensure strong mounting